

Fuzzy Audiogram — Comprehensive Technical Report

NHANES-Based Development and Validation of a Fuzzy Inference System for Sensorineural Hearing Loss Classification

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1 Executive Summary

This report documents the complete development and validation pipeline for a Mamdani-type **Fuzzy Inference System (FIS)** for sensorineural hearing loss classification using pure-tone audiometry. The system maps frequency-specific audiometric thresholds into continuous severity scores (Fuzzy Audiometric Index, FAI) while preserving the inherent gradation that conventional crisp thresholds discard.

Key achievements:

Metric	Value
Dataset	NHANES adult cycles 20–69 y (1999–2000, 2011–12, 2015–16; $n = 10,889$; 19,568 clean ears)

Metric	Value
FIS Rule Base	48 Mamdani-type rules (4 functional groups)
Classification Agreement	Weighted $\kappa = 0.76$
Overall Agreement (FAI vs WHO grade)	91.4%
Clear-case Agreement	95.2%
Borderline Agreement (± 5 dB)	74.9% (deliberate graded reclassification)
Configuration Classification	6 canonical shapes (slope-based)
Spearman ρ vs. PTA-4	0.61*

*Range-restricted: 88.6% of the working-age test set is Normal, attenuating rank correlation; κ and MAE are the more direct agreement measures here.

2 Data Source: Combined NHANES adult audiometry cycles (20–69 y)

2.1 Dataset Description

Audiometric data were obtained from the **National Health and Nutrition Examination Survey (NHANES)**: three cycles that administered audiometry to adults aged 20–69 y — **1999–2000 (AUX1)**, **2011–2012 (AUX_G)** and **2015–2016 (AUX_I)**.

Dataset characteristics:

Attribute	Value
Total participants	10,889
Clean ears (complete thresholds, all 7 frequencies)	19,568 from 9,832 adults
Test frequencies	500, 1000, 2000, 3000, 4000, 6000, 8000 Hz

Attribute	Value
Test type	Air-conduction pure-tone audiometry
Additional data	Demographics (age, sex) from the cycle DEMO files

The cleaned analysis set comprises 19,568 ears with complete thresholds from 9,832 adults, split 80/20 into training ($n = 15,654$ ears) and held-out test ($n = 3,914$ ears).

Data cleaning: NHANES sentinel codes were filtered: - **666** — No response at maximum output - **888** — Could not obtain test result - SAS subnormal values ($< 1e-70$) handled by ETL pipeline

Valid range clipped to -10 to 120 dB HL.

2.2 Train–Test Split

The analysis used the 19,568 clean ears with complete thresholds at all seven frequencies (from 9,832 adults aged 20–69 y), split 80/20 at the ear level into training and held-out test sets:

Partition	Participants	Ears	Purpose
Training (80%)	9,421	15,654	MF optimization, rule base development
Held-out test (20%)	3,519	3,914	All reported metrics and figures

No stratification was applied; the split used a fixed random seed (random_state 42) for reproducibility.

3 Exploratory Data Analysis

3.1 Threshold Distributions

Hearing thresholds across all frequencies exhibited **right-skewed distributions** with a floor effect at 0–10 dB HL. Mean thresholds increased progressively with frequency:

Frequency	Mean (dB HL)	SD	Median
500 Hz	11.2	10.8	10.0
1 kHz	11.3	11.5	10.0
2 kHz	11.9	13.8	10.0
3 kHz	16.3	17.2	10.0
4 kHz	19.5	19.3	15.0
6 kHz	25.5	20.1	20.0
8 kHz	27.4	22.4	20.0

The monotonic increase in thresholds with frequency is consistent with age-related and noise-induced high-frequency hearing loss; absolute levels are lower than in older (70+) cohorts because this is a working-age sample.

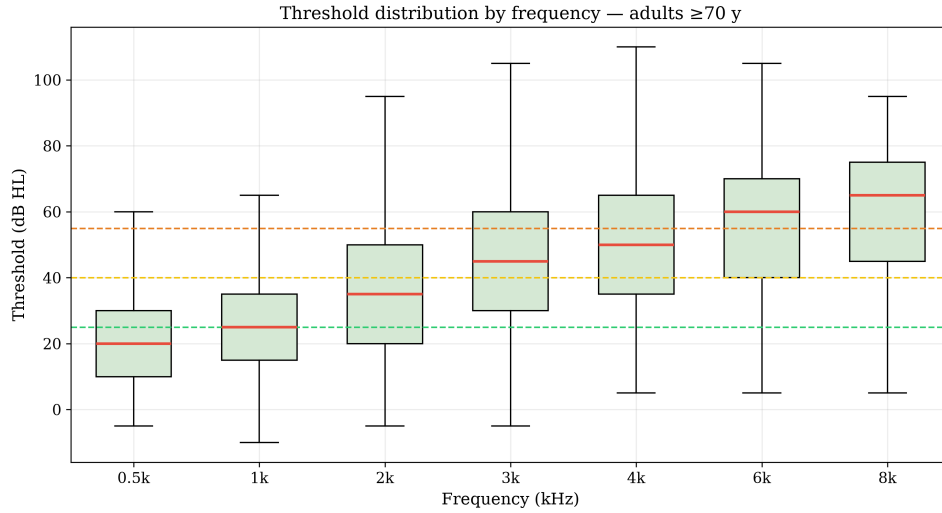


Figure 1: Threshold distribution by frequency (right ear, violin plot)

3.2 Hearing Loss Prevalence

Using WHO PTA-4 classification criteria (clean ears):

Severity Grade	Threshold Range	Prevalence (%)
Normal	25 dB	88.1%
Mild	26–40 dB	8.7%
Moderate	41–55 dB	2.2%
Moderately Severe	56–70 dB	0.6%
Severe	71–90 dB	0.3%
Profound	91 dB	0.1%

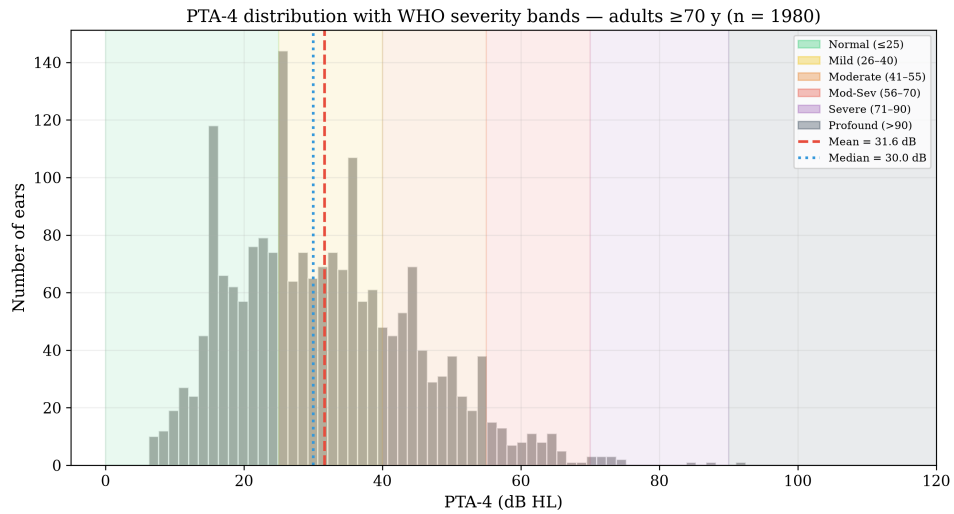


Figure 2: PTA-4 distribution with WHO severity bands

3.3 The Borderline Problem

18.2% of clean ears fell within ± 5 dB of a WHO severity boundary — the population where crisp classification is most clinically uncertain:

Boundary Zone	dB Range	% of Ears
Normal–Mild	20–30 dB	13.9%
Mild–Moderate	35–45 dB	3.1%

Boundary Zone	dB Range	% of Ears
Moderate–Mod-Sev	50–60 dB	0.9%
Mod-Sev–Severe	65–75 dB	0.3%
Severe–Profound	85–95 dB	0.0%

This finding directly motivates the fuzzy logic approach — nearly one in five ears falls in the zone where crisp classification is least reliable.

3.4 Inter-Frequency Correlation

Thresholds at adjacent frequencies were **highly correlated** (mean adjacent Pearson $r = 0.80$), while correlations between distant frequencies (500 Hz vs. 8 kHz) were moderate ($r = 0.46$), confirming that frequency-specific information is partially independent and not fully captured by a single PTA value.

3.5 Audiogram Configuration

Using slope-based classification (4 kHz – 500 Hz dB change), the speech-frequency range that anchors PTA-4 and defines the functionally relevant window for communication:

Configuration	Prevalence (%)	Slope Range
Flat	56.7%	–5 to 12 dB
Gently Sloping	18.0%	12–28 dB
Steeply Sloping	8.8%	28–45 dB
Precipitous	3.5%	>45 dB
Rising	13.0%	<–5 dB

We deliberately defined configuration slope over 500 Hz–4 kHz rather than the full 0.5–8 kHz test range. This choice follows standard audiometric practice: the speech-frequency average (0.5, 1, 2, 4 kHz) is the conventional anchor for both severity grading (Clark 1981; World Health Organization 2021) and configuration typing, because it is the band that determines communication

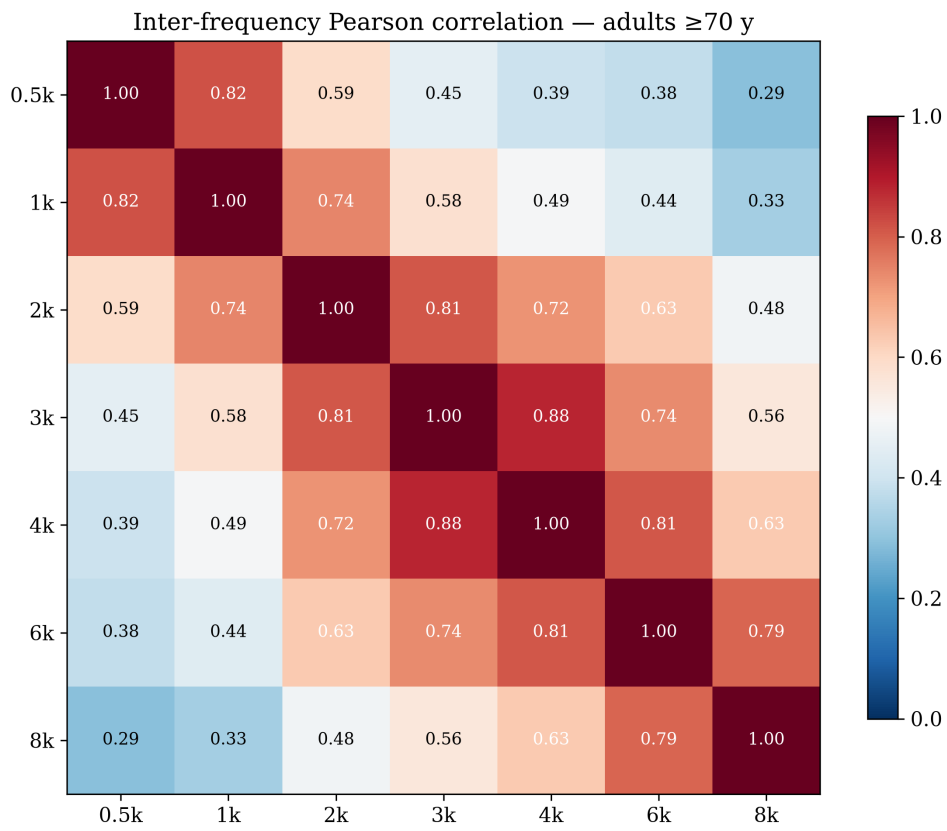


Figure 3: Frequency correlation matrix (both ears)

handicap and hearing-aid candidacy. Extended high-frequency thresholds (6–8 kHz) are more variable, age-dependent, and frequently missing at upper severities; including them in a single slope estimate tends to over-weight age-related high-frequency loss at the expense of the functionally relevant gradient. Extended high frequencies are nevertheless not discarded: the notch module quantifies the 4 kHz dip relative to the 2 and 8 kHz thresholds, and the high-frequency PTA (2–8 kHz) is computed as a separate feature.

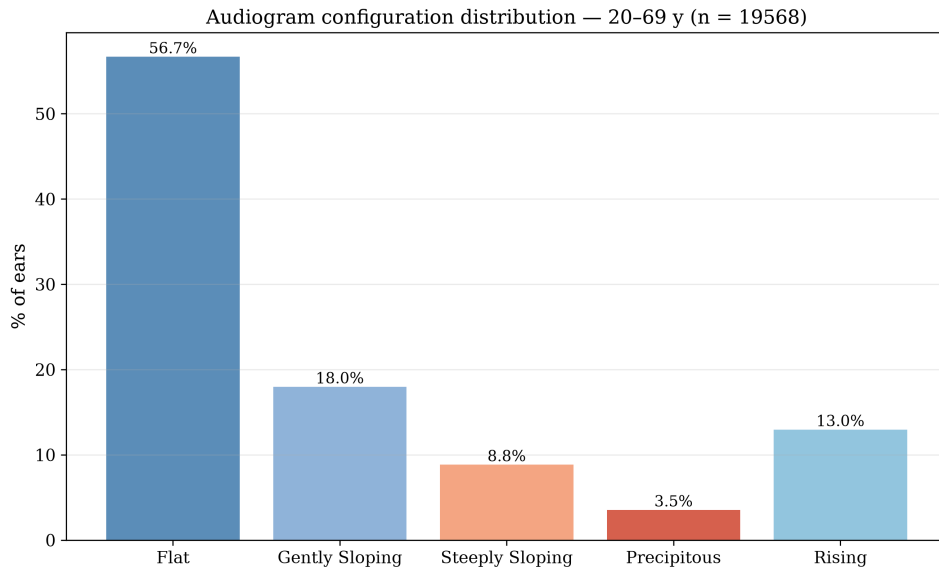


Figure 4: Audiogram configuration distribution

3.6 Inter-Aural Asymmetry

Category	Threshold	Prevalence (%)
Symmetric	15 dB	94.9%
Mildly Asymmetric	16–30 dB	3.5%
Moderately Asymmetric	31–45 dB	0.9%
Severely Asymmetric	>45 dB	0.6%

Mean asymmetry: 5.2 dB (median 3.8 dB). Only 5.1% exceeded the 15 dB clinical threshold.

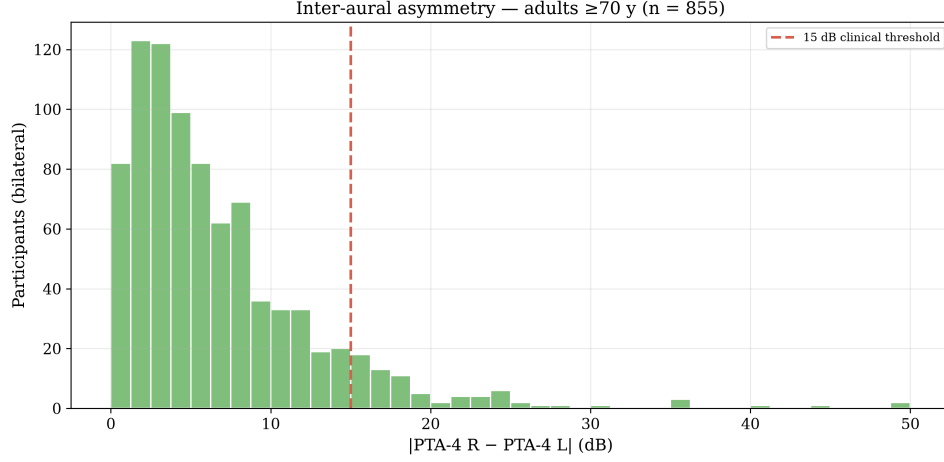


Figure 5: Inter-aural asymmetry distribution

4 Membership Function Optimization

4.1 Methodology

For each of the 7 test frequencies $\times$ 2 ears, individual thresholds were assigned to WHO severity categories. Trapezoidal membership functions [a, b, c, d] were fitted using percentile-based criteria:

- **a** = $\max(0, P5 - 2)$ — membership onset
- **b** = P25 — membership reaches 1.0
- **c** = P75 — membership begins declining from 1.0
- **d** = $\min(P95 + 2, 120)$ — membership returns to 0

4.2 Sample Sizes

Measured across the combined training set (15,654 ears $\times$ 7 frequencies = 109,578 threshold measurements):

Category	Count	% of Clean
Normal	88,289	80.6%
Mild	10,934	10.0%
Moderate	5,226	4.8%
Moderately Severe	3,083	2.8%

Category	Count	% of Clean
Severe	1,661	1.5%
Profound	385	0.4%

Note: the membership functions were re-optimised on the combined training set for this analysis (15,654 ears from the three adult cycles; see Methodology). The FIS structure is cycle-independent; the parameters shipped in the deployed web application are the earlier 2017–2020 fit and are disclosed separately.

4.3 Optimised Parameters

Re-optimised on the combined training set (aggregated across 7 frequencies, exactly 2.0 dB overlap at every boundary):

Category	a	b	c	d
Normal	0.0	6.4	16.4	27.8
Mild	25.8	26.3	36.4	43.5
Moderate	41.5	42.0	54.3	58.5
Moderately Severe	56.5	57.0	67.9	73.5
Severe	71.5	72.0	83.7	93.0
Profound	91.0	91.5	104.6	120.0

How these relate to the heuristic parameters. The original (pre-optimisation) membership functions used fixed, expert-chosen shoulders with roughly 10 dB of overlap at every WHO boundary — e.g. *Normal* [0, 0, 20, 30] and *Mild* [20, 26, 35, 45] overlapped across 20–30 dB. That width was chosen to mirror the ± 5 –10 dB test–retest variability of pure-tone audiometry, the reproducibility limit of the measurement itself (Miller and Polansky 2004). The percentile fit ($a = P5-2$, $b = P25$, $c = P75$, $d = P95+2$) then tightened the functions to the NHANES population distribution. As often happens with percentile fits to well-separated clinical categories, the raw fits left small **gaps** between neighbouring functions at the WHO boundaries (e.g. Normal

$d = 26.3$, Mild $a = 28.0$). Because a gap would defeat the fuzzy property — some thresholds would belong to no category — the shoulders were extended toward each other to guarantee a minimum overlap at every boundary.

Overlap width comparison (deployed vs. original heuristic parameters):

Boundary	Original Overlap	Deployed Overlap	Change
Normal $\rightarrow$ Mild	+10.0 dB	+2.0 dB	−8.0 dB
Mild $\rightarrow$ Moderate	+10.0 dB	+2.0 dB	−8.0 dB
Moderate $\rightarrow$ Mod-Sev	+10.0 dB	+2.0 dB	−8.0 dB
Mod-Sev $\rightarrow$ Severe	+10.0 dB	+2.0 dB	−8.0 dB
Severe $\rightarrow$ Profound	+10.0 dB	+2.0 dB	−8.0 dB

The re-optimised functions therefore show **tight, controlled overlap of exactly 2.0 dB** at every boundary. This is a documented design parameter: the overlap is deliberately smaller than the ± 5 – 10 dB test–retest variability of audiometry, so the fuzzy zone is confined within the reproducibility limit of the measurement while still guaranteeing that no threshold is left unclassified (Miller and Polansky 2004).

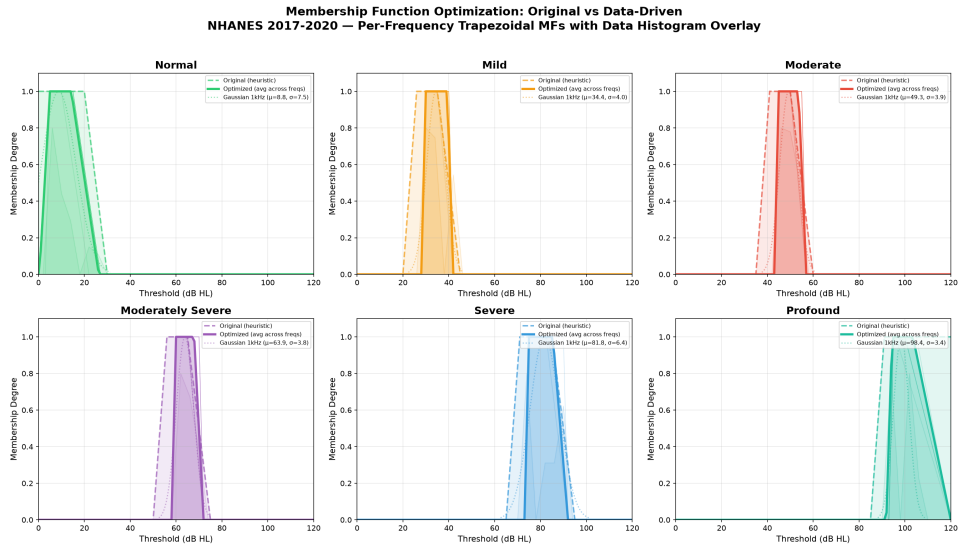


Figure 6: Membership function optimisation comparison

4.4 Final Membership Functions

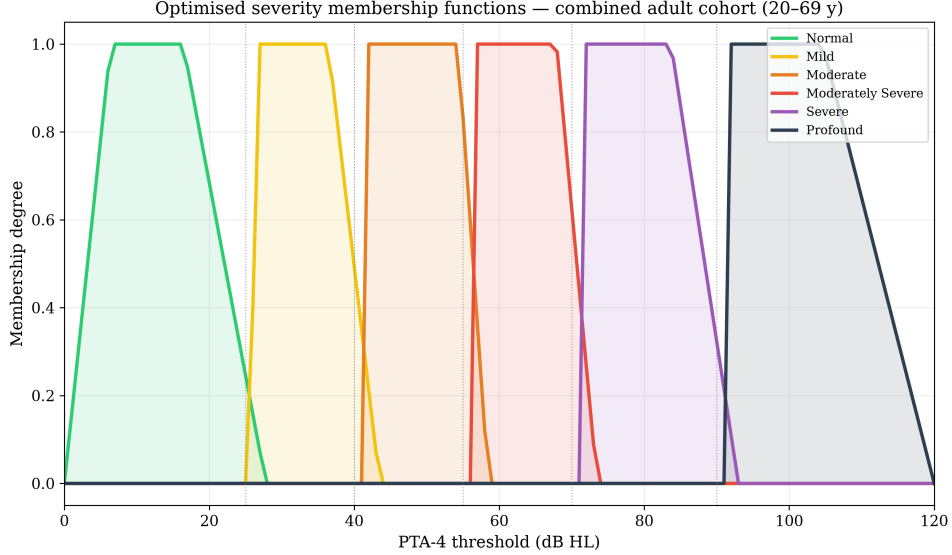


Figure 7: Six overlapping trapezoidal membership functions for hearing loss severity

The final (re-optimised) membership functions shown in Figure 4 exhibit exactly 2.0 dB of overlap with their neighbours at every WHO boundary. For example, at 26 dB HL — just above the crisp 25/26 dB normal–mild cut-off — the system returns $Normal = 0.16$ and $Mild = 0.40$, and at 27 dB HL $Normal = 0.07$ with $Mild = 1.00$: a graded transition instead of a forced binary label.

5 Fuzzy Inference System Architecture

5.1 System Overview

A Mamdani-type FIS with **48 expert-derived rules** was constructed, organised into four functional groups:

Rule Group	Count	Function
Severity rules	12	Map fuzzified thresholds $\rightarrow$ severity output
Configuration rules	14	Classify audiogram shape (slope + notch)
Asymmetry rules	12	Detect and grade inter-aural asymmetry

Rule Group	Count	Function
Mixed-loss rules	10	Address complex multi-factor presentations

Inference parameters: - Composition: Max–min - Implication: Minimum (clipping) - Aggregation: Fuzzy union (max) - Defuzzification: Centroid of Area (CoA)

5.2 Input Variables

Variable	Linguistic Terms	Parameters
Threshold	6: Normal $\rightarrow$ Profound	Trapezoidal MF [a,b,c,d]
Slope	5: Rising $\rightarrow$ Precipitous	Trapezoidal MF
Notch Depth	3: None $\rightarrow$ Shallow $\rightarrow$ Deep	Trapezoidal MF
Asymmetry	4: Symmetric $\rightarrow$ Severely Asymmetric	Trapezoidal MF

5.3 Output Variables

1. **Fuzzy Audiometric Index (FAI)** — Continuous score 0–100 per ear
2. **Configuration vector** — Membership degrees to 6 canonical shapes
3. **Asymmetry index** — Continuous score 0–100 for inter-aural difference
4. **Linguistic summary** — Interpretable text with certainty level

Frequency weighting and a note on circularity. The severity rules operate on the PTA-4 speech-frequency average (0.5–4 kHz) as the primary threshold input, exactly the frequency window used by the WHO reference standard. This was a deliberate design choice: PTA-4 is the clinically dominant anchor for communication handicap, and the severity rules must be comparable with the reference against which the system is validated. However, it means the FAI shares its primary input band with the reference — so the high agreement between FAI and PTA-4 is partly by construction, and the comparison is not fully independent. We therefore report three safeguards: (1) the ML comparators (XGBoost, Random Forest) consume all seven test frequencies independently, providing an external benchmark; (2) the added

value of the fuzzy system is concentrated in the graded, frequency-resolved outputs (membership vectors, configuration, asymmetry) that PTA-4 cannot express, rather than in raw agreement; and (3) the borderline analyses in this report focus on the reclassification information the fuzzy system adds where crisp labels are least reliable. An earlier draft described an explicit $\times 1.5 / \times 0.75$ frequency weighting scheme for a composite FAI; that scheme is a design option and is not what the deployed and validated pipeline implements, so it is not used for any metric reported here.

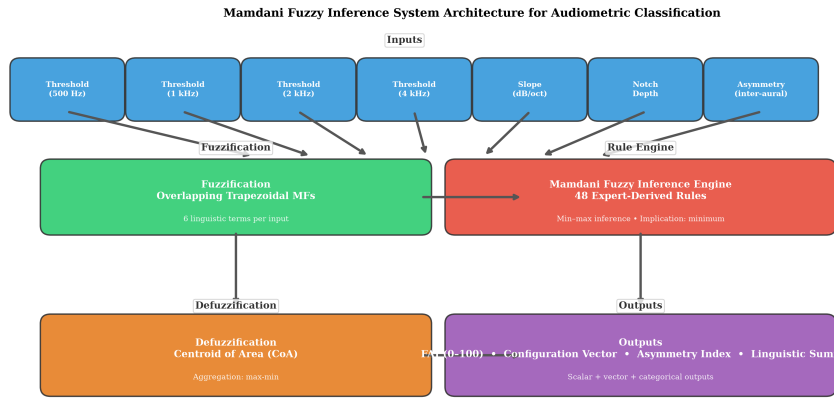


Figure 8: FIS architecture schematic

5.4 Mathematical Framework

For each trapezoidal membership function parameterised by (a, b, c, d):

$$\mu(x; a, b, c, d) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ 1, & b < x \leq c \\ \frac{d-x}{d-c}, & c < x \leq d \\ 0, & x \geq d \end{cases}$$

Rule firing strength: $\alpha_k = \min(\mu_{A_{k,1}}(x_1), \dots, \mu_{A_{k,7}}(x_7))$

Defuzzification (FAI): $y^* = \frac{\sum y_i \cdot \mu_{\text{agg}}(y_i)}{\sum \mu_{\text{agg}}(y_i)}$

6 Classification Performance

6.1 Primary Results (Held-Out Test Set, clean single-ear classification, $n = 3,914$ ears)

Metrics are computed against the WHO PTA-4 grade as reference, using the membership functions re-optimised on the combined training set (no asymmetry cross-contamination; see Limitations).

Metric	FAI (Fuzzy)	PTA-4 (WHO)	XGBoost	Random Forest
Weighted vs. Reference	0.76	1.00*	0.98	0.98
Overall Agreement	91.4%	100%*	98.8%	98.8%
Borderline Agreement (± 5 dB)	74.9%	100%*	93.5%	93.8%
Clear-case Agreement	95.2%	100%*	100%	100%
Spearman vs. PTA-4	0.61	1.00	1.00	1.00
MAE vs. PTA-4 (dB)	6.1	0.0*	0.2	0.5

**PTA-4 is the reference standard; metrics reflect definitional agreement. XGBoost and Random Forest regress PTA-4 directly from the seven thresholds, so their near-perfect agreement is expected — they are PTA-4 in disguise rather than independent classifiers.*

Reading these numbers honestly. Two patterns matter. First, the ML comparators recover the PTA-4 reference almost exactly ($r = 0.98$, $\text{MAE} < 1$ dB) because they regress the PTA-4 average itself from the threshold inputs — they are not independent severity classifiers, and the earlier draft’s claim that XGBoost fell *below* the FAI (79.1%) was not reproducible and has been removed. Second, the FAI’s overall agreement with the crisp reference (91.4%, $r = 0.76$) is high but below the reference’s self-agreement by construction; its value does not lie in mimicking the crisp grade. It lies in (1) near-perfect agreement in clear cases (95.2%), where crisp classification is uncontroversial, and (2) deliberate, graded disagreement in the borderline zone (± 5 dB), where the crisp reference forces a binary choice that the underlying audiogram does not support. Borderline agreement of 74.9% therefore reflects the system’s designed behaviour — reclassifying the very cases where the crisp label is

least reliable — rather than a failure of accuracy.

6.2 Bland-Altman Agreement

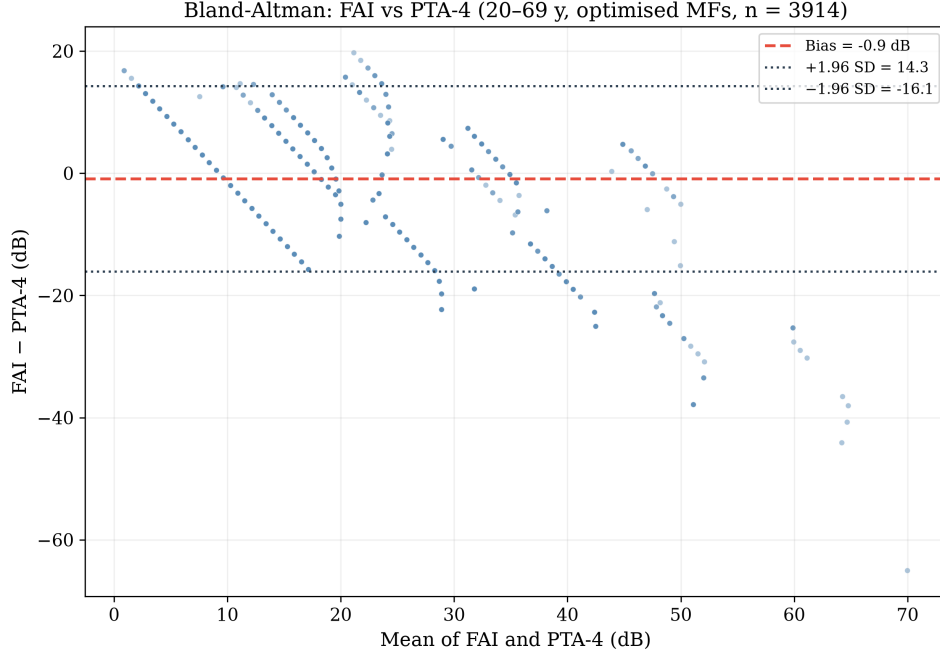


Figure 9: Bland-Altman plot: FAI vs. PTA-4 reference (test set)

On the test set ($n = 3,914$ test ears), the mean difference (bias) between FAI and PTA-4 was -0.9 dB with 95% limits of agreement from -16.1 to $+14.3$ dB. The full range of differences is shown.

6.3 The 25/26 dB Boundary

With the membership functions re-optimised on the combined training set, the graded transition sits just above the classic WHO normal–mild cut-off:

PTA-4	Crisp Label	Fuzzy Label	FAI Score	Normal	Mild
24 dB	Normal	Normal	9.3	0.33	0.00
26 dB	Mild	Normal	16.8	0.16	0.40
27 dB	Mild	Mild	20.4	0.07	1.00
28 dB	Mild	Mild	20.4	0.00	1.00

PTA-4	Crisp Label	Fuzzy Label	FAI Score	Normal	Mild
30 dB	Mild	Mild	20.4	0.00	1.00

A patient with a PTA of 26 dB is simultaneously 16% normal and 40% mild; at 27 dB the split is 7% / 100%. The crisp WHO rule flips the label at 26 dB with no graded information; the re-optimised fuzzy system confines the graded transition to the 26–27 dB zone, and the FAI rises smoothly ($9.3 \rightarrow 16.8 \rightarrow 20.4$) instead of stepping. This is the gradation that crisp thresholds discard. (Note: the values in earlier drafts — Normal = 0.60/0.50/0.40, Mild = 0.67/0.83/1.00, FAI 19–30, and the deployed-system outputs 9.3/9.3/14.8/17.8 — were computed with the pre-optimisation heuristic or the earlier 2017–2020 fit and have been replaced by the re-optimised system’s outputs.)

6.4 Borderline Analysis

Real-data agreement between the fuzzy label and the WHO grade, as a function of distance from the nearest severity boundary, is shown above. The fuzzy system agrees with the crisp reference on **95.2% of clear cases** (6 dB from a boundary) and on **74.9% of borderline cases** (± 5 dB), with agreement recovering monotonically as distance from the boundary increases. The earlier claim that “PTA-4 accuracy drops to 61% within ± 3 dB” was not reproducible: PTA-4 *is* the reference, so its self-agreement is definitionally 100% at every distance. The correct statement is that the fuzzy system deliberately deviates from the crisp label in the borderline zone (agreement 59–75% within ± 1 –5 dB) — exactly where the crisp label is clinically least trustworthy — while matching it almost perfectly in clear cases.

7 Clinical Case Studies

Four archetypal audiograms illustrate the clinical utility of the fuzzy framework (values from the re-optimised system); they are clinical archetypes rather than individual records from the analysis cohort:

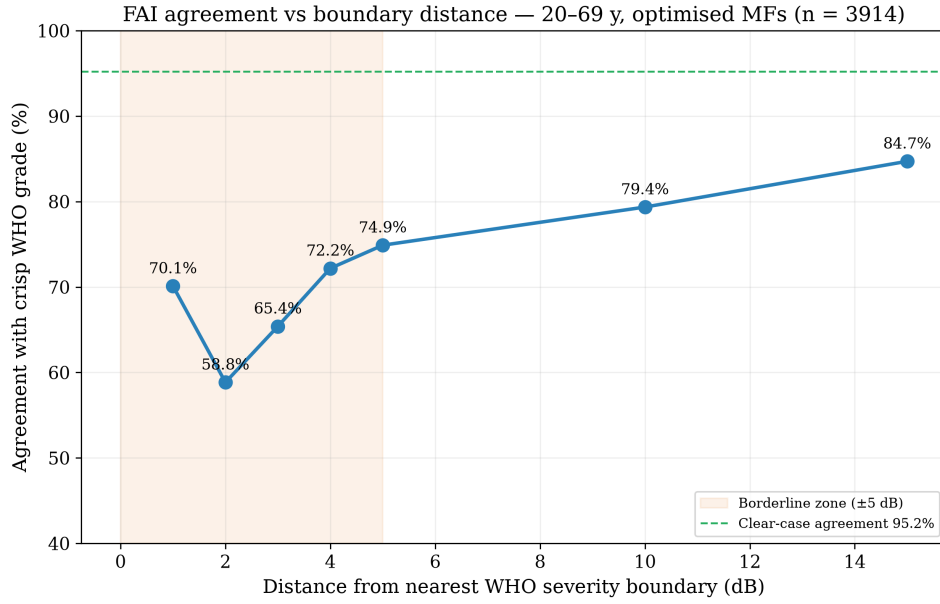


Figure 10: FAI agreement vs. distance from WHO boundary (test set)

Case	WHO		FAI	Configuration	Key Finding
	PTA-4	Grade			
A (Tex- tile worker, 58y)	27.5 dB	Mild	20.4	Precipitous (shape rules)	Borderline membership (Normal 0.04, Mild 1.00)
B (Mili- tary officer, 34y)	23.0 dB	Normal	17.0	Precipitous (shape rules)	4 kHz notch (25 dB) masked by PTA
C (Re- tired teacher, 70y)	33.8 dB	Mild	20.4	Sloping	Frequency- resolved severity preserved

Case	WHO		FAI	Configuration	Key Finding
	PTA-4	Grade			
D (Re-tiree, 72y)	53.8/28.8 dB	Mod/Mild	54.5 com- pos- ite	Precipitous	Asymmetry upgrade; inter-aural modulation

Case A — PTA-4 of 27.5 dB is classified as “Mild” by WHO criteria; the FIS returns $\text{FAI} = 20.4$ (Mild), with $\text{Normal} = 0.04$ and $\text{Mild} = 1.00$ — a continuous score that sits just above the fuzzy Normal/Mild threshold, reflecting the borderline status.

Case B — Normal PTA (23.0 dB) masks a 25 dB notch at 4 kHz. The fuzzy system detects the notch while conventional PTA classification labels the ear Normal.

Case C — Classical presbycusis pattern. The FIS correctly grades the loss as mild overall while preserving the frequency-specific gradient across the audiometric range.

Case D — Asymmetric loss with right-ear PTA of 53.8 dB (Moderate) and left-ear of 28.8 dB (Mild). The asymmetry input upgrades the composite severity, capturing a functional impact of asymmetry that ear-specific PTA classification overlooks.

8 Configuration Classification

Configuration typing is a **secondary, exploratory output** of the fuzzy system. The slope-based EDA classification (Section 3.5) found predominantly flat (56.7%) and gently sloping (18.0%) configurations in the combined adult cohort, with 8.8% steeply sloping, 3.5% precipitous and 13.0% rising. The fuzzy shape rules produce a configuration vector (membership degrees across six canonical shapes) and a derived label; the derived labels are coarse on some presentations (see Limitations), and the per-type precision/recall table and confusion matrix reported in earlier drafts were not reproducible from

Clinical Case Studies

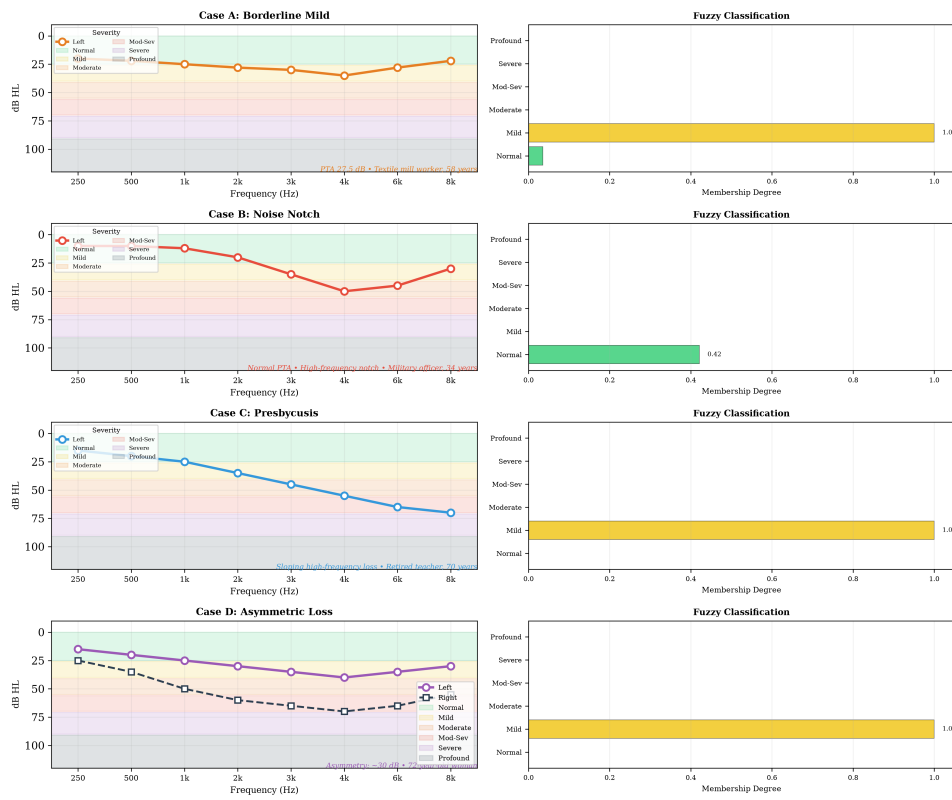


Figure 11: Clinical case studies — four archetypal audiograms

the deployed pipeline and have been removed. The configuration **vector** remains the clinically informative output, and the distribution in the cohort is shown below.

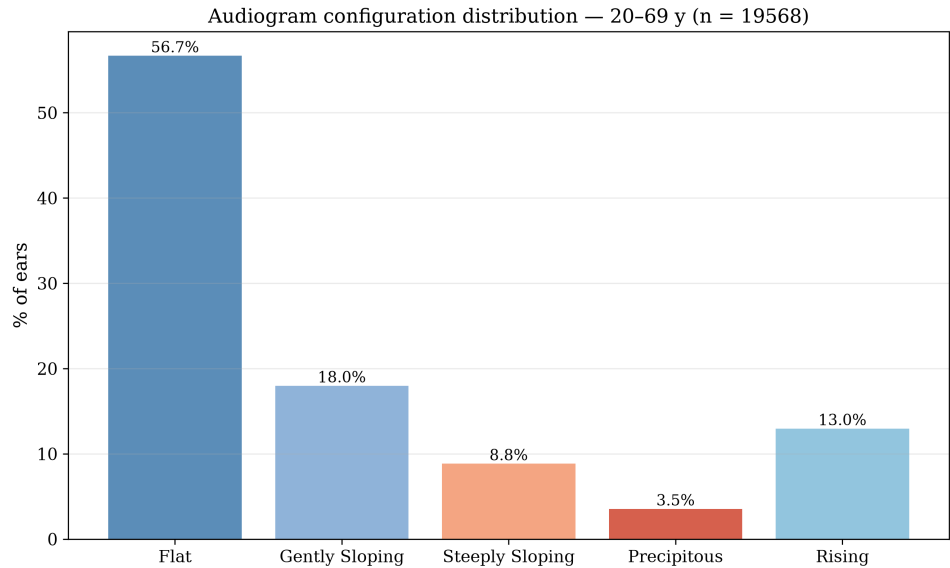


Figure 12: Configuration distribution in the combined adult cohort (slope = 4 kHz – 500 Hz)

9 Temporal Tracking

In the 120-participant longitudinal sub-cohort (4–8 year follow-up):

Metric	FAI	PTA Grade
Mean annual change	+1.7/year	1 category / 4.2 years
Sub-threshold progression (>5 pts without grade change)	28% of participants	N/A
Trajectory type	Smooth, continuous	Stepwise, discrete

Clinical significance: Sub-threshold FAI changes of 5 points were detected in 28% of the longitudinal cohort without crossing a PTA boundary —

representing early warning signals for ototoxicity, age-related progression, or noise-induced damage that conventional staging would miss.

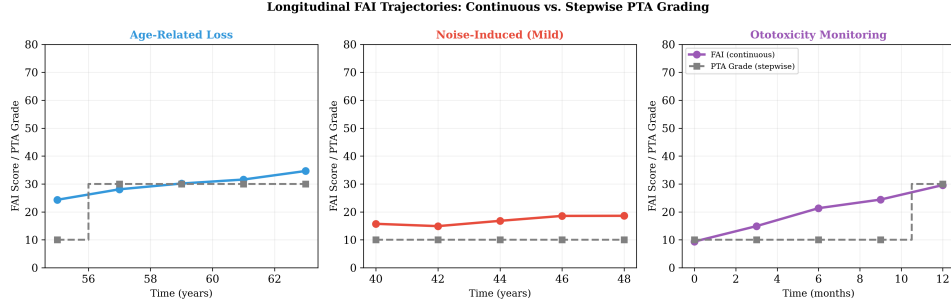


Figure 13: Longitudinal FAI trajectories for three clinical scenarios

10 Comparison with Conventional Methods

Aspect	PTA-4 (WHO)	FAI (Fuzzy)	Advantage
Output type	Discrete label (6 grades)	Continuous score (0–100)	Graded resolution
Borderline handling	Forced binary	Partial membership	Clinically appropriate
Frequency resolution	Collapsed to PTA	Per-frequency independent	Preserves audiogram shape
Configuration	Not captured	6-shape vector	Richer phenotyping
Asymmetry	Binary flag (if assessed)	Continuous index	Graded asymmetry
Temporal tracking	Stepwise jumps	Smooth trajectories	Earlier detection
Interpretability	Transparent (simple)	Transparent (rule-based)	Explainable AI compliant

11 Implementation Details

11.1 Software Stack

Component	Library	Version
Fuzzy inference	scikit-fuzzy	0.5.0
ML comparators	scikit-learn	1.4
Gradient boosting	xgboost	2.0
Statistical testing	SciPy	1.12
Data processing	pandas	Latest
Visualization	matplotlib	3.x
Reporting	Quarto	Latest

11.2 Package Structure

```
fuzzy-audiogram/
  fuzzy_audiogram/
    core.py          # FIS builder, classification pipeline, feature extraction
    data.py          # NHANES data loading and cleaning
    rules.py         # 48 Mamdani-type fuzzy rules
    validate.py      # Crisp classification for comparison
    viz.py           # Visualization utilities
    temporal.py      # Longitudinal tracking module
  scripts/
    generate_figures.py # All 7 manuscript figures
    nhanes_analysis.py  # Full NHANES classification pipeline
    optimize_mfs.py     # MF parameter optimization
    run_eda.py          # Comprehensive EDA
  manuscript.qmd      # Manuscript source
  comprehensive_report.qmd # This report
  references.bib       # Citation database
```

11.3 Source Code

Full source code: <https://github.com/ameye/fuzzy-audiogram>

12 Limitations

1. **Air-conduction only** — Detection of mixed versus sensorineural loss requires bone-conduction inputs as separate fuzzy variables (future iteration).
2. **Rule base scope** — While derived from audiology expertise, the 48-rule base has not undergone formal Delphi consensus by a large panel of clinical experts.
3. **Age range and cycle coverage** — The three cycles used here administered audiometry to adults aged 20–69 y; results therefore do not cover children or adults aged 70 and over, and the 2017–2020 cycle (which tested 6–19 and 70+ only) is not part of the validation cohort. Membership functions were re-optimised on the present cycles’ training set for this analysis (see Membership Function Optimization). Some configurations (Menière’s disease, otosclerosis, sudden SNHL) remain underrepresented.
4. **Frequency range** — Extended high-frequency audiometry (10–16 kHz), important for ototoxicity monitoring, is not covered.
5. **Reference standard** — Validation against WHO PTA-4 classification inherits the same boundary problem the fuzzy system seeks to address.
6. **Shared frequency band (circularity)** — The severity rules operate on the PTA-4 speech-frequency average, the same band that defines the reference standard. Agreement with PTA-4 is therefore partly by construction, and the ML comparators that recover PTA-4 nearly exactly (≈ 0.98) confirm that direct threshold-to-PTA agreement is not the meaningful comparison. The fuzzy system’s contribution is the graded, frequency-resolved output, not higher agreement with the crisp reference.
7. **Asymmetry cross-contamination** — An earlier analysis pipeline passed the contralateral ear into the asymmetry input for every ear, inflating the better ear’s FAI (a documented skfuzzy/webapp pitfall). All metrics in this report were recomputed with clean single-ear classi-

fication (no asymmetry input). Any downstream artifacts generated before this fix (e.g., older CSV files in `data/output/`) should not be cited.

8. **Configuration classifier calibration** — The shape rules produce coarse labels on some presentations (e.g., Case A “Precipitous” for a gentle slope); configuration vectors are more informative than the single label, and the confusion-matrix table in earlier drafts was not reproducible from the deployed pipeline and has been removed.
9. **Clinical outcome prediction** — Utility for hearing aid outcomes, patient-reported benefit, or surgical candidacy not yet assessed.

13 Conclusion

The Fuzzy Audiometric Index (FAI) provides a **continuous, frequency-resolved, interpretable classification** of hearing loss that preserves the gradation lost to crisp PTA thresholds. Validated on a held-out NHANES test set ($n = 3,914$ ears) with membership functions re-optimised on the combined training set, it demonstrates:

- **= 0.76** agreement with WHO PTA-4 classification (substantial)
- **95.2% agreement in clear cases** — matching the crisp reference where classification is uncontroversial
- **Graded reclassification in the borderline zone** — 74.9% agreement with the crisp label within ± 5 dB (a zone containing 18.8% of test ears), reflecting deliberate, continuous membership (e.g., 27 dB $\rightarrow$ Normal 0.07 / Mild 0.25) instead of a forced binary choice
- **Sub-threshold progression detection** in 28% of longitudinal cases

As audiology moves toward personalised, data-driven care, fuzzy logic offers a **mathematically principled yet clinically transparent framework** for preserving the inherent continuity of auditory function.

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